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Assessment of agroclimatology NASA POWER reanalysis datasets for temperature types and relative humidity at 2 m against ground observations over Egypt

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Abstract

The climatic reanalysis datasets are one of the most important data types that could help to overcome scarce of observations. Therefore, the main objective of this study is to evaluate NASA POWER reanalysis data for surface monthly average temperatures at 2 m (maximum (T_{max}), minimum (T_{min}), mean (T_{mean}) and dew point (T_d), all in °C) and relative humidity (RH) in percentage compared to the observed data at 20 Egyptian weather stations. The results showed that there are a significant correlation and goodness of fit between NASA POWER reanalysis and observed data for all parameters except RH. For temperature variables, the coefficient of determination (R^2) and Willmott Index of agreement (WI) attain around 0.75 and 0.90 respectively, while the root mean square error (RMSE) reaches to less than 5 °C; and the mean bias error (MBE) ranges from −3 to +3 °C for 85% of stations. In addition, NASA POWER accuracy of temperature parameters increases gradually northward with the highest ratio at the northern coast of Egypt. Where, it is slightly overestimated (under 3 °C) T_{max}, T_{min}, and T_{mean} at 80, 75 and 65% of the stations respectively. While it is slightly underestimated T_d with 3 °C at 90% of the stations. Contrarily, NASA POWER data accuracy of RH increases southward, particularly in Aswan. Additionally, NASA POWER reanalysis has a considerable underestimation for RH data at most stations, where the dominant MBE percentage ranges from −12 to −5% for about 75% of the stations. As well as, the maximum RMSE and MBE for all elements were recorded in the Malwi station at Middle Egypt. Consequently, R^2, WI, RMSE, and MBE for temperature parameters are almost within an acceptable range for most selected Egyptian stations which are located in the area that dominated by the influence of the Mediterranean Sea (northern of 30°N) and western of 30°E with a distance between 2.5 and 14.5 km from the Sea. While, the good RH estimation is recorded at stations that are located south of the area that dominated by the influence of the Mediterranean Sea (south of 26°N). Finally, NASA POWER reanalysis datasets can be used in case of missing or scarce of observations in Egypt. Nevertheless, it still needs improvements by taking into consideration the influence of the Mediterranean Sea and the locality of (especially middle) Egypt on temperature and in particular on relative humidity estimations.
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Keywords: NASA POWER reanalysis dataset; Egypt climatic zones; Statistical evaluations

1. Introduction

Climate and weather conditions can be regarded as the most significant factors affecting various human activities like agriculture, industry, environment etc. Although the

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meteorological satellites and ground stations underwent great development during the last few decades, most of the world countries still suffer from shortages in meteorological data, especially the developing countries. Therefore, the reanalysis and gridding meteorological data from global atmospheric models is considered as one of the most important weather and climate data sources, which can be used to compensate and filling the gaps and lacks measured and continuous meteorological data. The atmospheric reanalysis is produced from a mix of observations and model forecasts (Lindsay et al., 2013). There are numerous reanalysis datasets available over the globe for a long historical period, such as the National Center for Environmental Prediction/National Center for Atmospheric Research (NCEP/NCAR) reanalysis, the Climate Forecast System Reanalysis (CFSR), and the European Centre for Medium-Range Weather Forecasts (ECMWF). In addition, there are the Japanese Meteorological Agency Climate Data Assimilation System (JCDAS) reanalysis, Modern Era Retrospective-Analysis for Research and Applications (MERRA) by the National Aeronautics and Space Administration and NASA Prediction of Worldwide Energy Resource (NASA POWER) from the Goddard Earth Observing System (GEOS) reanalysis. Although the different reanalysis datasets could use in several fields of applications, it is necessary to investigate and evaluate its accuracy by comparing it against the observations.

NASA's has long supported satellite systems and research providing data, including long-term estimates of meteorological quantities and surface solar energy fluxes, for the climate and climate processes studying (Stackhouse et al, 2004). According to NASA POWER data access service (https://power.larc.nasa.gov/docs/v1/), this data is available as a single point, regional, and global coverage with daily, interannual, and climatology temporal average. Where, the single point endpoint produces a time series of data based on the registered coordinate (only single latitude and longitude). As well as, the regional endpoint gives a time series based on a bounding box of latitude and longitude coordinates. While the global endpoint returns long-term climatological averages for the entire globe. The available meteorological variables are average of near surface air temperature (maximum, minimum, mean and dew point), relative humidity, precipitation, solar radiation and wind speed and direction. It can be used for many purposes such as climatic variability studies and crop modeling applications (Monteiro et al., 2018).

Many previous kinds of research evaluated some reanalysis data over the globe. El Afandi (2014) performed a comparison between the Climate Forecast System Reanalysis (CFSR) and the ground-based observations for 23 stations in Egypt. He used 2-meter maximum, minimum, mean and dew point temperatures, as well as mean sea level pressure and relative humidity for the period from January 1979 to December 2006. His results proved that CFSR reanalysis data gave a good agreement with the observations at most stations. He also pointed out that, the CFSR is acceptable in a good manner to be used in cases of a lack of observed meteorological data from ground stations, but it needs some improvements at fewer stations. Furthermore, many studies compared and evaluated most of parameters from various reanalysis datasets such as the NCEP/NCAR, ECMWF, MERRA, GLDAS and Japanese 25-year Reanalysis (JRA-25) products (e.g., Flocas et al., 2005; Bosilovich et al., 2007; Wanqiu et al., 2010; Keller et al., 2011; Decker et al., 2012; Zib et al., 2012; Bao and Zhang, 2013; Kaicun and Robert, 2013; Lindsay et al., 2013; Schade et al., 2013; Sharp et al., 2015; Jones et al., 2016; Jones et al., 2016; Yang and Zhang, 2017) against numerous sources of observation like satellite, onshore and offshore synoptic weather stations and Radio Sound based on varied climatic periods over different regions of the globe. The majority of these studies proved that most of the reanalysis datasets gave an acceptable accuracy with a great matching and in good harmony with the observations.

On the other hand, the NASA POWER datasets have few studies that investigated the applicability of the NASA POWER dataset for different applications especially agriculture sector across the globe (Jha et al., 2018). Bai et al. (2010) tested and verified the NASA satellite and model-derived weather dataset from the NASA POWER database for daily maximum and minimum temperatures and total solar radiation, to complete the shortage of the monitored data from the ground stations in China. Their results proved that there is coincide between NASA POWER and the monitored data with acceptable accuracy for handling in China. White et al. (2011) compared daily solar radiation data from the NASA POWER with instrument readings from 295 stations and values estimated by Weather Generator for Solar Radiation generator in the continental United States. They concluded that NASA POWER solar data are a promising resource for studies requiring realistic accounting of historic variation. In addition, White et al. (2008) compared daily temperature from NASA POWER for 1983 to 2004 over the continental US for 855 individual ground stations from the National Weather Service Cooperative Observer Program. They found that NASA POWER data need some improvements by adjusting for elevation (lapse rate) effects, reducing seasonal bias, and refining estimation of actual maximum and minimum temperatures in diurnal cycles. Furthermore, Wart et al. (2013) evaluated the capacity of NASA POWER along with other two gridded weather databases to simulate crop yield potential or water-limited yield potential of rice in China, USA maize, and wheat in Germany. Their results revealed that the simulated crop yield potential or water-limited yield potential using observed daily weather data combined with solar radiation from the NASA POWER database was in much better agreement with control weather data with little bias. The robust and evaluation of NASA POWER database for daily near surface minimum, maximum and average air temperature, solar radiation, rainfall, relative humidity and wind speed

was carried out through its comparison with 302 Brazilian ground weather stations, to quantify its impacts as a data source on potential and attainable sugarcane yields (Monteiro et al., 2018). They proved that NASA POWER products gave better results at higher latitudes and elevation and could be used as source of climatic data for agricultural activities with reasonable confidence for regional and national spatial scales, although it still need improvements to wind speed and RH estimation. Negm et al. (2017) assessed the suitability of NASA POWER archive to estimate daily meteorological variables (minimum, maximum, and average air temperature, relative humidity, global solar radiation and wind speed) and reference evapotranspiration in Sicily, Italy, for the period 2006–2014, based on ground data measured by a network of climate stations belonging to the regional Agrometeorological Information Service. Their result showed that NASA POWER agroclimatology database is proper to obtain the suitable estimations of daily evapotranspiration, even to other Mediterranean countries where most of the climate variables are not measured. However, there are inaccurate estimations of relative air humidity occurring for the coastal weather stations as well as of the air temperature for those inland stations characterized by high elevations. Also, daily air temperatures retrieved from the NASA POWER archive were proposed and validated for the climate of Sicily, Italy (Negm et al., 2018). Where, the ground elevation from Digital Elevation Model was used to estimate the spatial and temporal variability of a lapse rate adjustment factor used to downscale and to disaggregate POWER-NASA air temperature. The downscaled daily air temperature results were consistent and comparable with local ground measurements. Finally, NASA POWER reanalysis dataset for different atmospheric parameters over Egypt has not evaluated. So the main objective of this study is to investigate and evaluate NASA POWER reanalysis data for surface monthly average of 2 m (maximum, minimum, mean and dew point) air temperature (°C) and relative humidity (%) compared to the observed data at 20 Egyptian meteorological stations distributed along the different climatic zones in Egypt.

1.1. Study area

Egypt is located between latitudes from 22°N to 32°N and longitudes from 25°E to 37°E, as shown in Fig. 1. The geography of Egypt is unique because it is located between the Mediterranean Sea from the north and the Red Sea and Gaza strip from the east, as well as it bounded by Sudan from the south and Libya from the west.

The desert covers more than 90% of Egypt, which surrounds the majority of the Egyptian land. This geographical location makes Egypt exposed globally and locally to different climatic features throughout the different year seasons. It is characterized by the Mediterranean climate in the north to be close to the desert and semi-desert climate in the south (El Afandi et al., 2013). It is hot, dry in summer, and temperate in winter with irregular rainfall (Domrose and El-Tantawi, 2005).

2. Data and methodology

Monthly mean observations of the selected 20 ground weather stations were obtained from the Egyptian Meteorological Authority (EMA) for the period of 1983–2006. The chosen atmospheric parameters are 2-meter maximum (T_{max}), minimum (T_{min}), mean (T_{mean}), dew point (T_d) temperature °C and relative humidity % (RH). The geographical location and information of these stations are shown in both Fig. 1 and Table 1.

The same weather parameters for the same period of observations were collected from NASA POWER daily basis dataset with a horizontal resolution of 1° latitude by 1° longitude, which available on NASA's website http://power.larc.nasa.gov/. In addition, the single point technique (endpoint) is used to fitting the selected weather station sites. These data converted to a monthly average for the selected parameters to be compatible with the observations. Finally, the simple kriging interpolation method was applied to produces a good spatial interpolation for the annual average of temperature, dew point and relative humidity over Egypt using the Geographical Information System (ArcGIS-10.3) software.

There were many missing data from the ground weather observations, as shown in Table 1. Therefore, the corresponding data from the NASA dataset has been deleted to ensure the compatibility with the observational one. In addition, to enhance the quality of the results, all unacceptable data as negative values are removed.

2.1. Statistical procedures for data evaluation

The statistical procedure is the most important tool to investigate the goodness of fit and accuracy of the predicted or estimated data compared to the measured one. Therefore, different statistical methods using Microsoft Excel have been applied and executed between the monthly NASA POWER data and measurements. These statistical measurements are Coefficient of Determination (R^2), Willmott Index of agreement (WI), Root Mean Square Error (RMSE) and Mean Bias Error (MBE).

2.1.1. Coefficient of determination (R^2)

It is a statistical measurement, which shows how much variation in the observed data explained by the variation in the predicted one (Pineiro et al., 2008). It ranges from 0 to 1, where higher values indicate less error variance and that the model has a good fit, and where typically values greater than 0.5 are considered acceptable (Santhi et al., 2001; Van Liew et al., 2003; Moriasi et al., 2007).

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(O_i - R_i)^2}{\sum_{i=1}^{n}(O_i - O_a)^2} \quad (1)$$

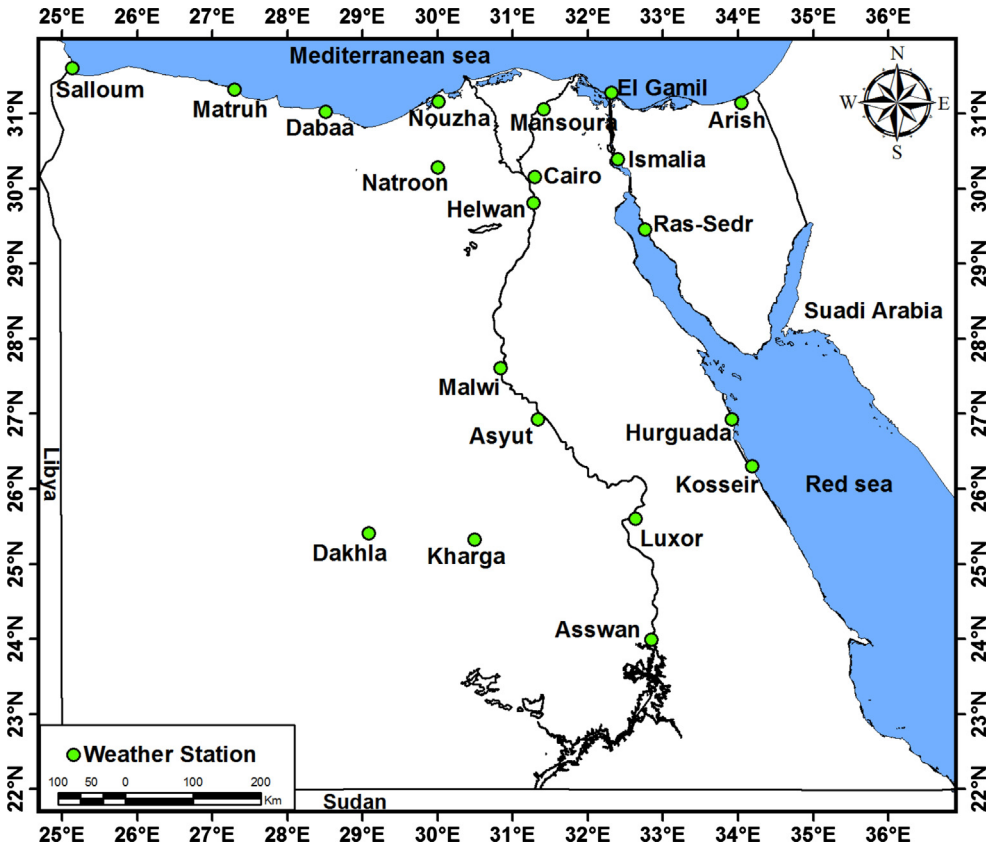


Fig. 1. Egypt domain includes the selected weather stations used in this study.

where O_i, O_a and R_i represent the observed, observed average and reanalysis values respectively and n is the number of the observations available for analysis.

2.1.2. Willmott index of agreement (WI)

It is the standardized measure of the degree of model prediction error, which varies between 0 and 1. A value of 1 indicates a perfect match, and the value of 0 indicates no agreement at all (Willmott 1981). The Willmott Index of agreement (WI) developed by Willmott (1981) to overcome the insensitivity of Nash-Sutcliffe efficiency (E) and coefficient of determination (R^2) to differences in the measured and predicted means and variances (Legates and McCabe, 1999; Krause et al., 2005).

$$WI = 1 - \frac{\sum_{i=1}^{n}(O_i - R_i)^2}{\sum_{i=1}^{n}(|R_i - O_a| - |O_i - O_a|)^2} \quad (2)$$

2.1.3. Root mean Square Error (RMSE) and its normalization (NRMSE)

Root Mean Square Error (RMSE) represents the mean standard deviation of model prediction with respect to the observation (Kobayashi and Salam, 2000; Gauch et al., 2003; Pineiro et al., 2008; Bai et al., 2010).

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}(R_i - O_i)^2}{n}} \quad (3)$$

The Normalized RMSE (NRMSE) values obtained by dividing RMSE by the average observed (Mentaschi et al., 2013; Sarkalkan et al., 2015; Taebi and Mansy, 2017) such as:

$$NRMSE = \frac{RMSE}{O_a} * 100\% \quad (4)$$

where lower RMSE and NRMSE values indicate higher model accuracy and less residual variance (Jamieson et al., 1998).

2.1.4. Mean bias Error (MBE) and its normalization (NMBE)

The Mean Bias Error (MBE) is a measure of systematic error between the predicted and observed values (Singh et al., 2013).

$$MBE = \frac{\sum_{i=1}^{n}(R_i - O_i)}{n} \quad (5)$$

In addition, it is normalized by dividing its values by the observed average (NMBE) (Mentaschi et al., 2013; Sayad et al., 2015).

$$NMBE = \frac{MBE}{O_a} * 100\% \quad (6)$$

The MBE and NMBE measure of the model is over or under estimation with its positive or negative values respectively.

Table 1
Characteristics and descriptions of the selected 20 Egyptian ground weather stations used in comparison.

Station name	Latitude (°N)	Longitude (°E)	Elevation (m)	Distance (km) from the nearest point of		Available observations	Missing observations
				Mediterranean sea	Red sea		
Salloum	31.57	25.08	189	8.92	733.68	164	124
Matruh	31.33	27.22	23	2.87	530.80	288	0
Dabaa	30.93	28.47	45	14.5	404.51	288	0
Nouzha	31.2	29.95	−1	3.26	282.24	288	0
El Gamil	31.28	32.23	3	0.04	151.24	288	0
Arish	31.03	33.8	98	35.72	184.12	288	0
Mansoura	31	31.45	5	57.54	155	288	0
Ismalia	30.6	32.25	15	76.18	76.5	288	0
Natroon	30.4	30.2	1	97.79	225.35	288	0
Cairo	30.13	31.4	68	156.93	106.66	288	0
Helwan	29.72	31.2	50	209.92	116.64	288	0
Ras-Sedr	29.58	32.72	4	162.12	0.77	288	0
Malwi	27.7	30.75	47	418.72	271.77	264	24
Asyut	27.05	31.02	229	503.43	277.77	288	0
Hurghada	27.18	33.8	16	440.15	1.88	288	0
Kosseir	26.13	34.25	30	559.4	2.08	262	26
Luxor	25.67	32.7	87	656.23	181.94	288	0
Dakhla	25.48	29	113	616.22	561.85	208	80
Kharga	25.45	30.53	87	672.07	413.09	288	0
Aswan	23.97	32.78	182	841.84	290.33	288	0

3. Results and discussion

3.1. Annual climate pattern of Egypt

The annual climate for the selected atmospheric variables is calculated from the observed monthly mean data (T_{max}, T_{min}, T_{mean}, T_d, and RH) at each station, during the period from 1983 to 2006 as shown in Table 2. Then, ArcGIS software is used to interpolate these annual climates between stations to cover all Egypt, as illustrated in Fig. 2. This figure shows that the range of annual average temperatures are from 23.11° to 32.12 °C for T_{max}, from 11.72° to 20.29 °C for T_{min}, from 19.13° to 26.42 °C for T_{mean} and from 4.43° to 15.3 °C for T_d, while RH ranges from 27.41% to 70.61%. The annual spatial distributions of the selected climatic elements in Egypt are mainly affected by the astronomical and geographical location (especially latitudes), distance from water bodies, and the contrast in the levels of the earth's surface (elevations above sea level). The Mediterranean Sea on the north coast of Egypt plays a significant and vital role in most of Egypt's climate throughout the year. Where, the Mediterranean Sea covers a large area and the currents of wind coming from the north above the Mediterranean pass to the land of Egypt without any obstructed from high mountain barriers. Most of Egypt's land is flat, so the Mediterranean Sea has a remarkable impact on meteorological elements patterns and their distributions particularly the northern part of Egypt up to 30° N. South of 30°N the influence of the Mediterranean Sea is rapidly decreasing until it has almost become no longer after Middle Egypt (i.e., after latitude 26°N). On the other hand, the influence of the Red Sea is limited and is concentrated especially along its parallel coasts and mountains, because it is surrounded by a high range of mountains, in addition to the prevailing winds throughout the year is the North West wind, and then they are parallel to the Red Sea coast.

Table 2
Annual climate of T_{max}, T_{min}, T_{mean}, T_d and RH at the selected stations.

Station name	T_{max} °C	T_{min} °C	T_{mean} °C	T_d °C	RH %
Salloum	24.48	14.42	19.13	11.75	67.26
Matruh	24.25	15.33	19.70	13.48	68.73
Dabaa	24.89	15.31	19.94	13.32	67.14
Nouzha	24.65	16.11	20.30	14.10	68.73
El Gamil	23.11	17.36	21.23	15.30	69.39
Arish	25.05	13.51	19.82	13.65	69.83
Mansoura	26.06	13.22	19.98	13.82	70.61
Ismalia	27.04	14.78	21.69	12.70	60.28
Natroon	27.16	14.24	21.01	11.32	57.88
Cairo	26.56	15.79	21.82	12.12	57.87
Helwan	26.81	15.05	21.59	10.94	54.62
Ras-Sedr	26.99	15.38	22.02	12.04	55.80
Malwi	27.40	11.72	20.62	12.47	64.60
Asyut	28.36	14.11	22.20	6.45	40.26
Hurghada	28.39	18.62	24.56	9.98	41.98
Kosseir	26.79	20.29	24.56	13.81	56.66
Luxor	32.12	15.66	24.80	8.92	39.89
Dakhla	30.99	14.29	23.57	6.99	37.86
Kharga	31.64	15.86	25.00	8.80	39.67
Aswan	31.96	18.57	26.42	4.43	27.41

Therefore, it is obvious from Fig. 2 and Table 2 that, the highest annual T_{max} and T_{mean} was recorded in the inland stations and the lowest annual average recorded in the coastal stations, because it is not affected by marine influences. In addition, the effect of the Mediterranean Sea on

the average annual relative humidity is also evident, as it is high in the coastal stations and low in the internal stations. Where, T_{max} and T_{mean} increase gradually from the Northwest to the Southeast, on the contrary with T_d and RH. Furthermore, T_{min} in the western desert is less than that in the eastern desert and most of Mediterranean coast due to the specific heat capacity of water and land that leads to heating loses from the desert land at night faster than the water surfaces that consequently affects the coastal parts. Finally, it is clear that, there is an inverse relationship between average annual temperature and relative humidity over Egypt due to the Mediterranean Sea impacts.

3.2. Evaluation of NASA POWER maximum temperature

Coefficient of determination (R^2), Willmott Index of agreements (WI), Root Mean Square Error (RMSE °C), Normalized Root Mean Square Error (NRMSE %), Mean Bias Error (MBE °C) and Normalized Mean Bias Error (NMBE %) are used to evaluating monthly average of NASA POWER maximum temperature at the selected stations based on actual observations as illustrated in Table 3 to the left.

Tables 1 and 3 indicate that NASA POWER success to simulate maximum temperature at latitudes north of 30°N (area that dominated by the influence of the Mediterranean Sea) that have longitudes western of 30°E. in addition, it is provide good results at the stations that have distance between 2.5 and 14.5 km from Sea with all elevations. Where, R^2 and WI are highest with the lowest RMSE at the coastal stations that have longitudes western of 30°E. While R^2 and WI are lowest with highest RMSE at stations south of 30°N or east of 30°E with distance from Sea less than 2.5 km or larger than 15 km. The maximum values of R^2 (0.962) and WI (0.989) are registered at Matruh while the lowest values (0.317 and 0.712 respectively) are observed at Kosseir. Also, the RMSE ranges from 0.979 °C with NRMSE 4.037% in Matruh to 7.405 °C with NRMSE 27.030% in Malwi. As well as, the MBE ranges from −1.451 °C with NMBE −5.568% in Mansoura to 2.170 °C with NMBE 8.101% in Kosseir. Furthermore, the maximum temperature (T_{max}) reanalysis data is slightly higher than the observations at most of the selected weather stations where the bias is positive at 14 of 20 stations (70%).

The WI values reached to more than 0.75 for most stations and more than 50% of them have R^2 over 0.5 (Fig. 3a). On the other hand, about 85% of the stations have RMSE values less than 7 °C (Fig. 3d) with NRMSE less than 25% (Fig. 3b). In addition, 90% of the stations have a limited difference between approximately −2.0 °C and + 2.0 °C (Fig. 3c); associated with NMBE nearly between −6 and +6% (Fig. 3b). Lastly, four coastal stations (Salloum, Matruh, Dabaa, and Nouzha) have a high correlation and goodness of fit with the NASA POWER T_{max} dataset, as shown in Fig. 3. Whereas the accuracy of NASA POWER at other stations varies according to its geographical location (south of 30°N and east of 30° E) and distance from the Sea (less than 2.5 km or larger than 15 km).

Finally, the average R^2, WI, RMSE, and MBE for all stations is 0.564, 0.830, 5.230 and 0.499 respectively. Consequently, this proves relative goodness of fit and agreement between the grounds T_{max} observations with the NASA POWER dataset at most of the selected Egyptian stations.

3.3. Evaluation of NASA POWER minimum temperature

Table 3 to the right demonstrates the statistical indicators for the monthly average of minimum temperature (T_{min}) from NASA POWER against observations from weather stations. One may notice that the maximum and minimum value of R^2 (0.971 and 0.512) is recorded at Matruh and El-Gamil stations respectively. The WI ranges from 0.792 in Mansoura to 0.961 in Nouzha, the RMSE ranges from 1.955 °C with NRMSE 12.135% in Nouzha to 5.9 °C with NRMSE 50.359% in Malwi. In addition, the MBE ranges from −2.720 °C with NMBE 14.608% in Hurghada to 4.709 °C with NMBE 40.194% in Malwi. Although R^2 for T_{min} is more than 0.50 at all stations, NASA POWER T_{min} dataset register highest accuracy at stations that located at latitudes north of 30°N and have longitudes western of 30°E and distance between 2.5 and 14.5 km from Sea with the different elevations (same as for T_{max}). These conditions were proper at Salloum, Matruh, Dabaa and Nouzha stations, so they have better results than the other stations as shown in Fig. 4

Furthermore, NASA POWER reanalysis dataset for T_{min} have 70% positive mean bias (overestimation) compared to measurements, while 30% of stations with a negative bias are located in North Egypt. Therefore, NASA POWER T_{min} values are higher than the observations of most weather stations in Egypt. In addition, 85% of the stations have a difference that ranges between −3.0 and +3.0 °C (Fig. 4c) with NMBE between −20% and +20% (Fig. 4b). On the other hand, more than 90% of these stations have a RMSE less than 5 °C (Fig. 4d) with NRMSE 25% or less (Fig. 4b), and most of their R^2 and WI are higher than 0.75 (Fig. 4a). Generally, the average of R^2, WI, RMSE and MBE at all stations attains to 0.739, 0.877, 3.995 and 1.297 respectively. These values indicate that there are an agreement and a relatively high correlation between NASA POWER T_{min} data and the observed one for most selected Egyptian stations.

3.4. Evaluation of NASA POWER mean temperature

The monthly average of mean temperature from NASA POWER at all selected stations are at a higher fitness than both minimum and maximum temperature compared to the observation, as shown in Table 4 to the left. It is clear from Table 4 to the left and Fig. 5 that R^2 at all stations is

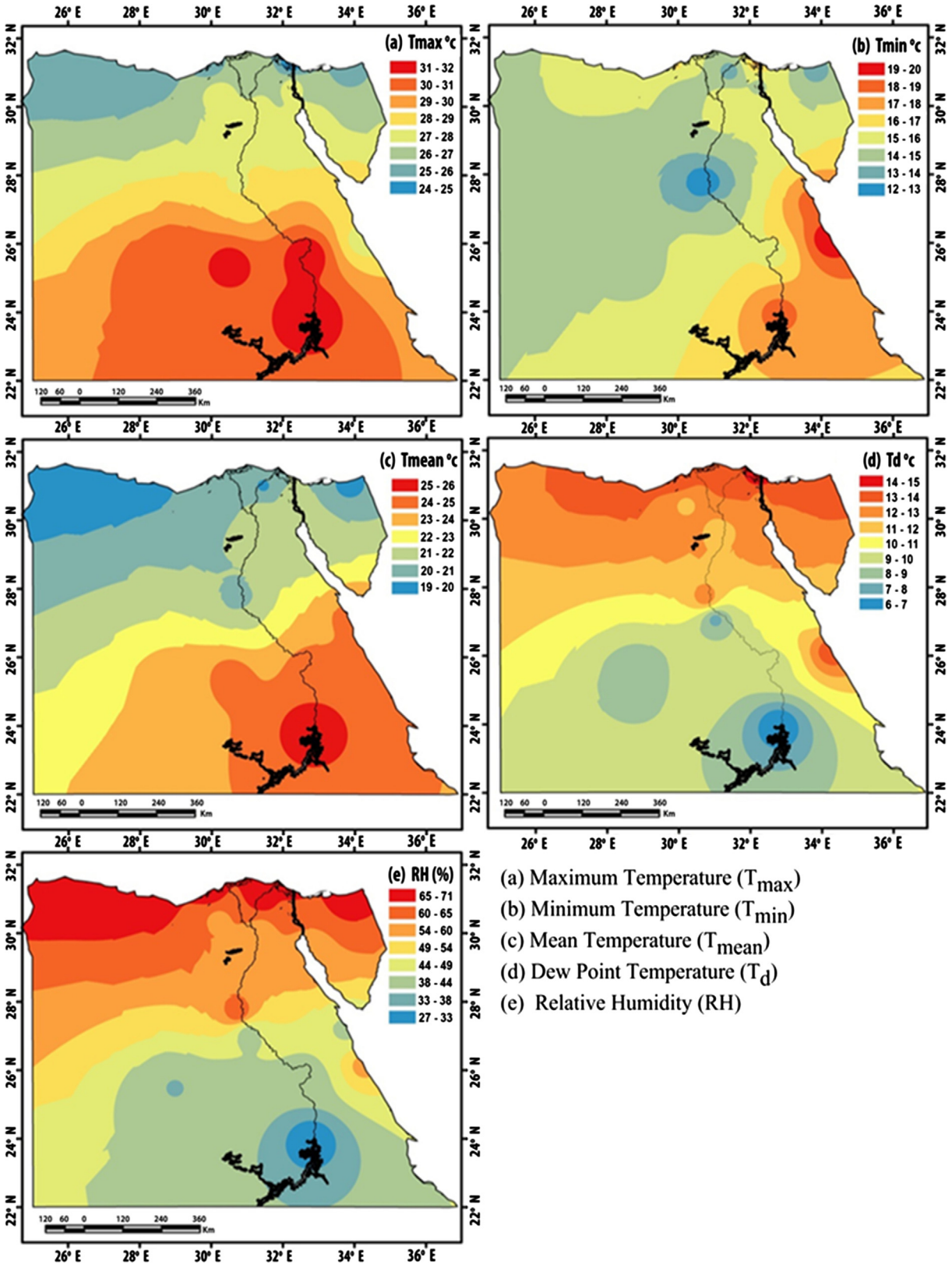


Fig. 2. Annual climate of (a) T_{max}, (b) T_{min}, (c) T_{mean}, (d) T_d and (e) RH over Egypt during the period from 1983 to 2006.

larger than 0.95 with maximum value at Kosseir station (0.987) and minimum value at Mansoura station (0.944) with an average of 0.971 over all stations. Also, WI values are highest than 0.94 and range from 0.944 in Salloum to 0.995 in Dakhla with an average of 0.98. In addition, the RMSE ranges from 0.956 °C in Nouzha to 2.537 °C in

Table 3
Statistical indicators for monthly average of maximum and minimum temperature.

Station	Maximum Temperature						Minimum Temperature					
	R-Square	WI	RMSE	NRMSE %	MBE	NMBE %	R-Square	WI	RMSE	NRMSE %	MBE	NMBE %
Salloum	0.955	0.986	1.193	4.874	0.209	0.854	0.969	0.851	3.999	27.727	3.884	26.932
Matruh	0.962	0.989	0.979	4.037	0.196	0.808	0.971	0.910	2.881	18.794	2.758	17.988
Dabaa	0.950	0.985	1.147	4.609	−0.382	−1.534	0.968	0.908	2.957	19.310	2.795	18.253
Nouzha	0.940	0.983	1.211	4.911	0.040	0.162	0.960	0.961	1.955	12.135	1.452	9.012
El-Gamil	0.384	0.752	5.629	24.356	1.393	6.027	0.512	0.822	4.313	24.841	−0.468	−2.697
Arish	0.452	0.764	5.383	21.494	−1.136	−4.534	0.692	0.827	4.427	32.762	3.132	23.182
Mansoura	0.507	0.775	5.729	21.983	−1.451	−5.568	0.723	0.792	5.275	39.901	4.223	31.944
Ismalia	0.514	0.819	5.710	21.119	0.650	2.403	0.696	0.895	3.530	23.883	−1.297	−8.776
Natroon	0.526	0.822	5.814	21.410	1.490	5.489	0.684	0.902	3.423	24.038	1.088	7.640
Cairo	0.523	0.818	5.817	21.899	1.700	6.401	0.646	0.891	3.640	23.058	−0.712	−4.508
Helwan	0.528	0.823	5.953	22.207	1.708	6.370	0.669	0.905	3.479	23.109	0.095	0.628
Ras-Sedr	0.500	0.819	5.840	21.640	0.584	2.163	0.677	0.869	4.097	26.648	−2.165	−14.078
Malwi	0.388	0.758	7.405	27.030	2.129	7.771	0.725	0.823	5.900	50.359	4.709	40.194
Asyut	0.522	0.824	6.206	21.885	1.073	3.785	0.769	0.912	3.918	27.766	2.122	15.034
Hurghada	0.421	0.788	6.541	23.040	1.893	6.667	0.639	0.848	4.974	26.712	−2.720	−14.608
Kosseir	0.317	0.712	6.202	23.148	2.170	8.101	0.517	0.818	4.722	23.275	1.583	7.805
Luxor	0.477	0.797	6.936	21.598	−0.719	−2.238	0.749	0.904	4.238	27.071	2.076	13.260
Dakhla	0.461	0.792	6.876	22.185	−0.969	−3.127	0.770	0.909	4.134	28.937	2.293	16.052
Kharga	0.477	0.792	6.947	21.958	−1.374	−4.343	0.760	0.922	3.841	24.217	1.094	6.898
Aswan	0.486	0.798	7.073	22.133	0.766	2.395	0.691	0.902	4.194	22.583	−0.005	−0.027

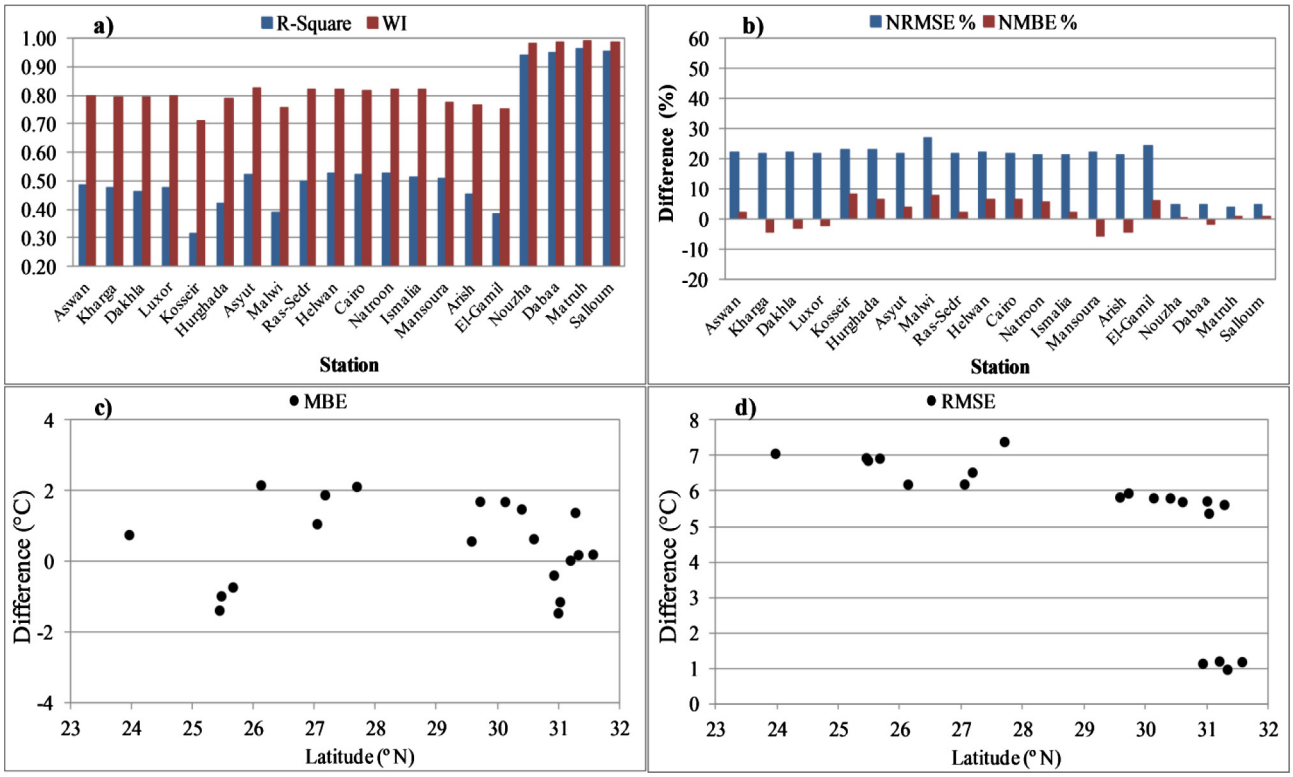


Fig. 3. Statistical indicators for monthly average of maximum temperature.

Malwi with an average of 1.477; while the NRMSE ranges from 4.711% to 12.304% respectively with an average value of 6.811%. At the same time, the MBE ranges from −1.856 °C in Ras-Sedr to 2.204 °C in Malwi with NMBE from −8.427% to 10.687% respectively.

It is clear from Fig. 5a that R^2 and WI values are over 0.95 for 95% of the selected stations. However, all stations have RMSE less than 3 °C (Fig. 5d) and 85% of them have NRMSE of 10% or less (Fig. 5b). 55% of the stations have a positive MBE (overestimation), which means that the T_{mean} NASA POWER is higher than measurements at most stations. In addition, 90% of these stations have a difference between −2.0 and 2.0 °C (Fig. 5c) with NMBE almost less than ±10% (Fig. 5b). So, it is indicated that

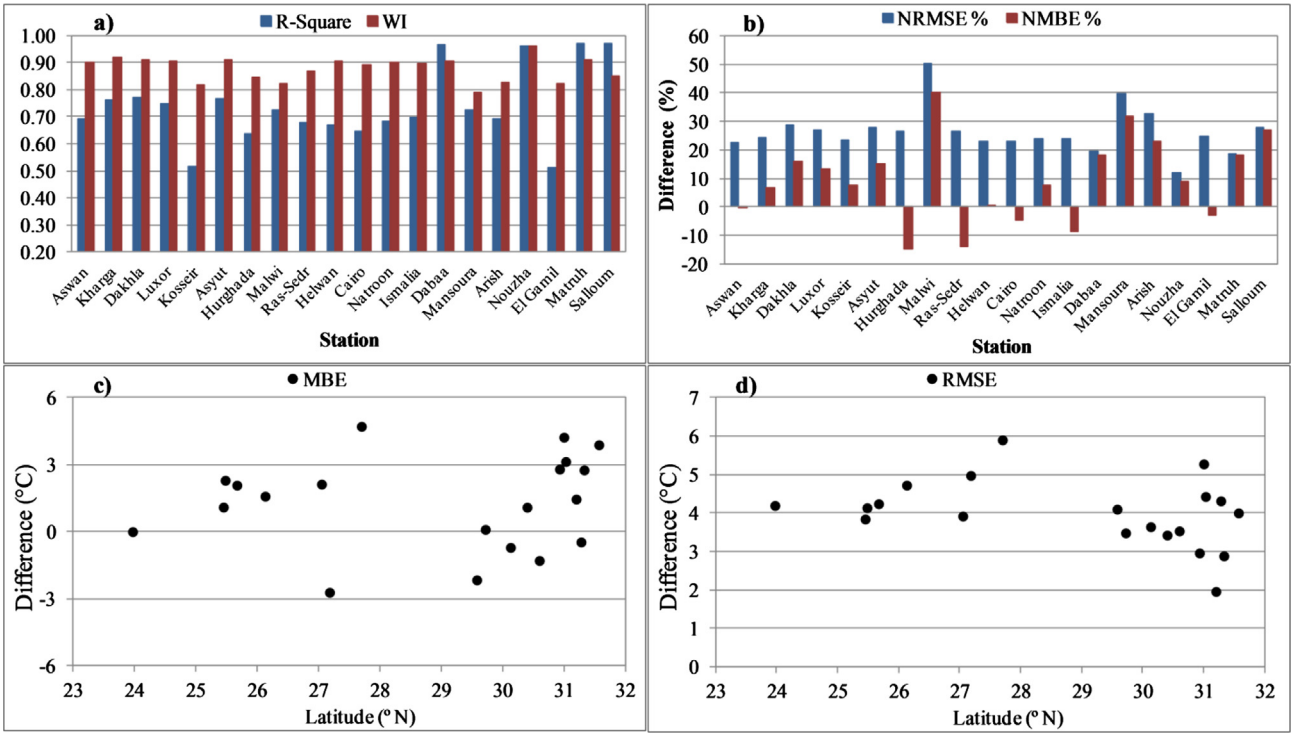


Fig. 4. Statistical indicators value for monthly average of minimum temperature.

there is an agreement, high correlation, and goodness of fit between NASA POWER and observation dataset of T_{mean} at all selected stations.

Generally, regarding to temperature variables (T_{max}, T_{min}, and T_{mean}), NASA POWER simulates T_{max} and T_{min} with highest accuracy at stations that located northern of 30°N (area that dominated by the influence of the Mediterranean Sea) and western of 30°E with distance between 2.5 and 14.5 km from the Sea at all different elevations. These conditions were performed at Salloum, Matruh, Dabaa and Nouzha stations. On the contrary, T_{mean} from NASA POWER is more accurate with the highest agreement and goodness of fit compared to the observations at all stations with very little under and over estimations at most stations. Comparable results of NASA POWER's temperature variables estimations were found by Bai et al. (2010) in China, Monteiro et al. (2018) in Brazil and Negm et al. (2017; 2018) in Sicily, Italy. In spite of the approval of those results, they concluded that the accuracy of NASA POWER mainly depends on the stations' elevation. However, this impact is not significantly evident in the Egyptian stations used in this study since their elevations are below 100 m except for only four stations (Salloum, Asyut, Dakhla and Aswan).

3.5. Evaluation of NASA POWER dew point temperature

The maximum values of R^2 (0.978) and WI (0.991) and the minimum value of RMSE (0.890 °C) and NRMSE (6.312%) are recorded in Nouzha while the minimum values of R^2 (0.651) and WI (0.612) and the maximum value of RMSE (7.804 °C) and NRMSE (62.577%) are in Malwi, as pointed out in Table 4 to the right. NASA POWER at the stations located north of 29.7°N (the area that dominated by the influence of the Mediterranean Sea) has the highest accuracy with best values of R^2 and WI as well as smallest values of RMSE except at Natroon station which also still has acceptable values. Moreover, The MBE ranges from −7.201 °C with NMBE of −57.736% in Malwi to 1.344 °C with NMBE of 11.443% in Salloum.

Statistical indicators R^2 and WI reach to above 0.75 for 75% of the stations, as shown in Fig. 6a. Furthermore, their RMSE is less than 5 °C (Fig. 6d) with NRMSE less than 20% for 40% of the stations that locate in the northern part of Egypt (Fig. 6b). 90% of these stations have a negative MBE (underestimation); which implies that NASA POWER reanalysis data for T_d is lower than the measurements at most stations. Also, 90% of the stations have a difference between −6.0 °C and 0 °C (Fig. 6c); with NMBE mostly under 20% for 50% of stations and above 20% at the remaining 50% of stations (Fig. 6b).

Generally, the average values of R^2, WI, RMSE and MBE for all stations reach to 0.855, 0.860, 3.153 and −2.395 respectively. Hence, NASA POWER reanalysis T_d is highly correlated and in good agreement with the observations, particularly in the northern part of Egypt.

3.6. Evaluation of NASA POWER relative humidity

From Table 5 and Fig. 7, one may notice that the reanalysis relative humidity (RH) from NASA POWER does not have a good correlation with the observations.

Table 4
Statistical indicators for monthly average of mean and dew point temperature.

Station	Mean Temperature						Dew Point Temperature					
	R-Square	WI	RMSE	NRMSE %	MBE	NMBE %	R-Square	WI	RMSE	NRMSE %	MBE	NMBE %
Salloum	0.974	0.944	2.375	12.412	2.183	11.409	0.932	0.959	1.911	16.267	1.344	11.443
Matruh	0.977	0.973	1.529	7.760	1.344	6.825	0.945	0.983	1.225	9.089	−0.080	−0.596
Dabaa	0.969	0.977	1.425	7.146	1.130	5.667	0.972	0.988	1.016	7.629	0.279	2.098
Nouzha	0.974	0.990	0.956	4.711	0.523	2.577	0.978	0.991	0.890	6.312	−0.468	−3.319
El-Gamil	0.957	0.981	1.287	6.063	−0.827	−3.897	0.967	0.951	2.063	13.482	−1.861	−12.165
Arish	0.959	0.985	1.179	5.947	0.172	0.870	0.968	0.988	1.066	7.813	−0.488	−3.577
Mansoura	0.944	0.973	1.654	8.277	0.783	3.919	0.964	0.988	0.997	7.213	−0.079	−0.570
Ismalia	0.951	0.968	2.001	9.222	−1.571	−7.241	0.934	0.863	3.697	29.123	−3.492	−27.506
Natroon	0.985	0.992	0.974	4.636	0.476	2.268	0.801	0.919	2.720	24.022	−1.315	−11.615
Cairo	0.973	0.989	1.161	5.321	−0.564	−2.585	0.950	0.935	2.432	20.066	−2.148	−17.723
Helwan	0.976	0.992	1.036	4.799	0.018	0.082	0.921	0.865	3.415	31.208	−3.166	−28.935
Ras-Sedr	0.957	0.963	2.229	10.124	−1.856	−8.427	0.887	0.738	5.461	45.350	−5.238	−43.500
Malwi	0.966	0.962	2.537	12.304	2.204	10.687	0.651	0.612	7.804	62.577	−7.201	−57.736
Asyut	0.984	0.993	1.065	4.797	0.638	2.873	0.816	0.919	2.323	36.017	−1.458	−22.607
Hurghada	0.976	0.976	1.905	7.755	−1.471	−5.987	0.802	0.645	6.090	61.009	−5.828	−58.377
Kosseir	0.987	0.989	1.030	4.194	0.762	3.103	0.862	0.935	2.097	15.187	−1.239	−8.972
Luxor	0.975	0.993	1.162	4.687	−0.183	−0.736	0.705	0.632	5.636	63.164	−5.269	−59.057
Dakhla	0.982	0.995	1.006	4.266	−0.286	−1.212	0.699	0.760	4.074	58.311	−3.444	−49.289
Kharga	0.982	0.984	1.716	6.864	−1.401	−5.604	0.608	0.647	5.866	66.643	−5.239	−59.516
Aswan	0.977	0.991	1.305	4.941	−0.646	−2.447	0.743	0.876	2.273	51.347	−1.504	−33.970

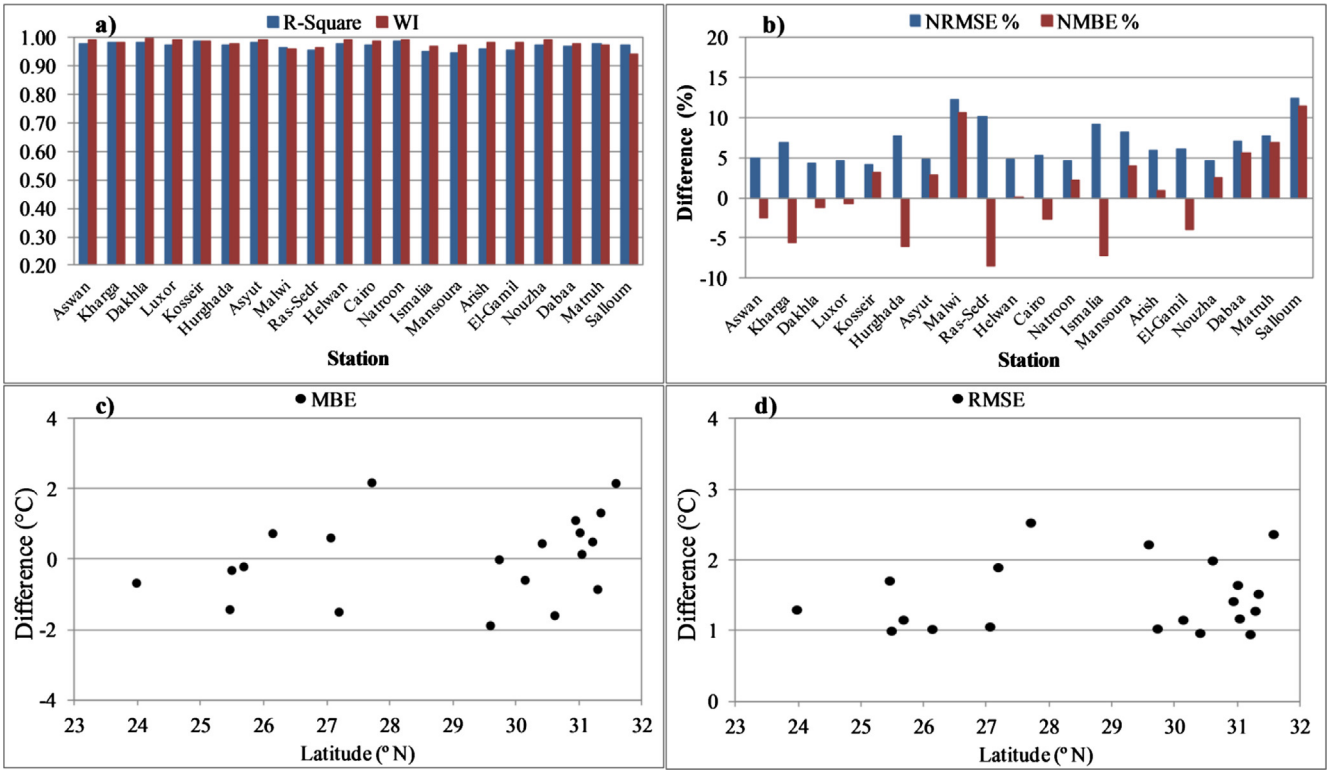


Fig. 5. Statistical indicators value for monthly average of mean temperature.

The maximum and minimum values of R^2 achieves 0.902 and 0.003 are recorded in Aswan and Kosseir respectively. The WI values range from 0.326 at El-Gamil to 0.938 in Aswan. In addition, the RMSE ranges from 4.182% with NRMSE of 15.258% in Aswan to 31.749% with NRMSE of 49.145% in Malwi. In addition, the MBE ranges from −2.81% with NMBE of −10.254% in Aswan to −30.34% with NMBE of −46.956% in Malwi. It is noted that the best statistical indicator values are registered at the stations located southern of 26°N (south of area that dominated by the influence of the Mediterranean Sea) such as Aswan, Luxor, Dakhla and Kharga stations. While statistical

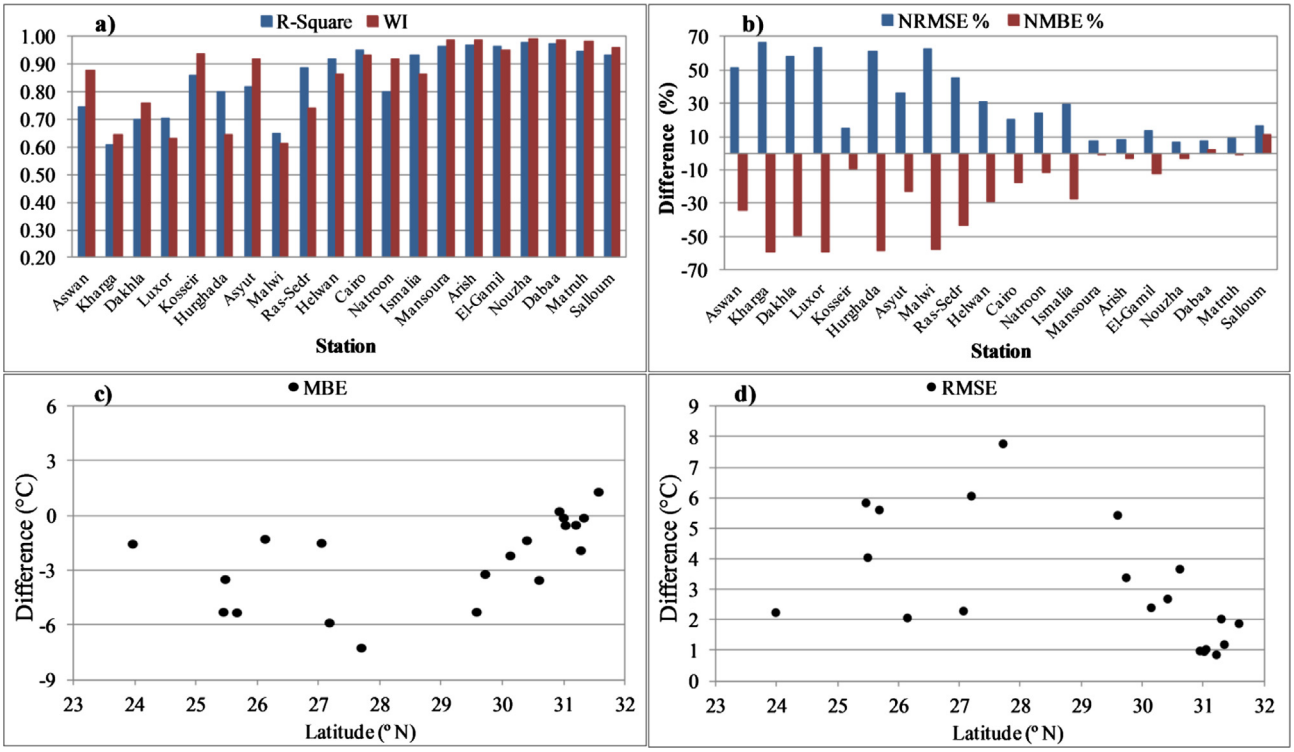


Fig. 6. Statistical indicators for monthly average of dew point temperature.

Table 5
Statistical indicators for monthly average of relative humidity.

Station	R-Square	WI	RMSE	NRMSE %	MBE	NMBE %
Salloum	0.017	0.443	11.556	17.183	−7.190	−10.691
Matruh	0.011	0.384	10.863	15.805	−6.837	−9.947
Dabaa	0.045	0.380	8.601	12.810	−4.588	−6.833
Nouzha	0.016	0.351	8.271	12.033	−5.068	−7.373
El-Gamil	0.011	0.326	9.065	13.064	−4.788	−6.901
Arish	0.037	0.400	7.917	11.339	−4.669	−6.687
Mansoura	0.105	0.481	9.004	12.752	−6.244	−8.844
Ismalia	0.351	0.603	11.504	19.085	−8.903	−14.770
Natroon	0.243	0.588	12.133	20.963	−8.187	−14.146
Cairo	0.454	0.658	9.644	16.664	−7.607	−13.144
Helwan	0.701	0.612	12.426	22.752	−11.491	−21.039
Ras-Sedr	0.438	0.533	13.228	23.705	−11.551	−20.699
Malwi	0.340	0.395	31.749	49.145	−30.335	−46.956
Asyut	0.826	0.838	7.778	19.317	−6.658	−16.536
Hurghada	0.694	0.629	11.647	27.744	−10.719	−25.533
Kosseir	0.003	0.387	19.474	34.369	−11.599	−20.471
Luxor	0.870	0.690	12.893	32.320	−12.216	−30.622
Dakhla	0.786	0.773	9.062	23.935	−8.103	−21.401
Kharga	0.704	0.703	11.908	30.019	−10.451	−26.347
Aswan	0.902	0.938	4.182	15.258	−2.810	−10.254

indicator values are unacceptable at those stations that located into the area that dominated by the little influence of the Mediterranean Sea (northern of 26°N) and also have distance from the Sea less than 2.5 km like Kosseir and Nouzha stations.

Fig. 7 reveals that about 20% of the stations have R^2 and WI greater than 0.75 (Fig. 7a) with nearly 50% of them have RMSE less than 10% (Fig. 7d) and NRMSE less than 20% (Fig. 7b). While, all of these stations have a negative MBE (underestimation) which indicates that the reanalyzed RH is lower than the measurements at all stations. In addition, 75% of the stations have a difference between −12.0 and −5.0% (Fig. 7c) with NMBE less than 25% at most stations (Fig. 7b).

Generally, the average values of R^2, WI, RMSE, and MBE at all stations attain 0.378, 0.556, 11.645 and −9.001 respectively. Therefore, the reanalysis RH over Egypt has a small correlation and does not agree with most

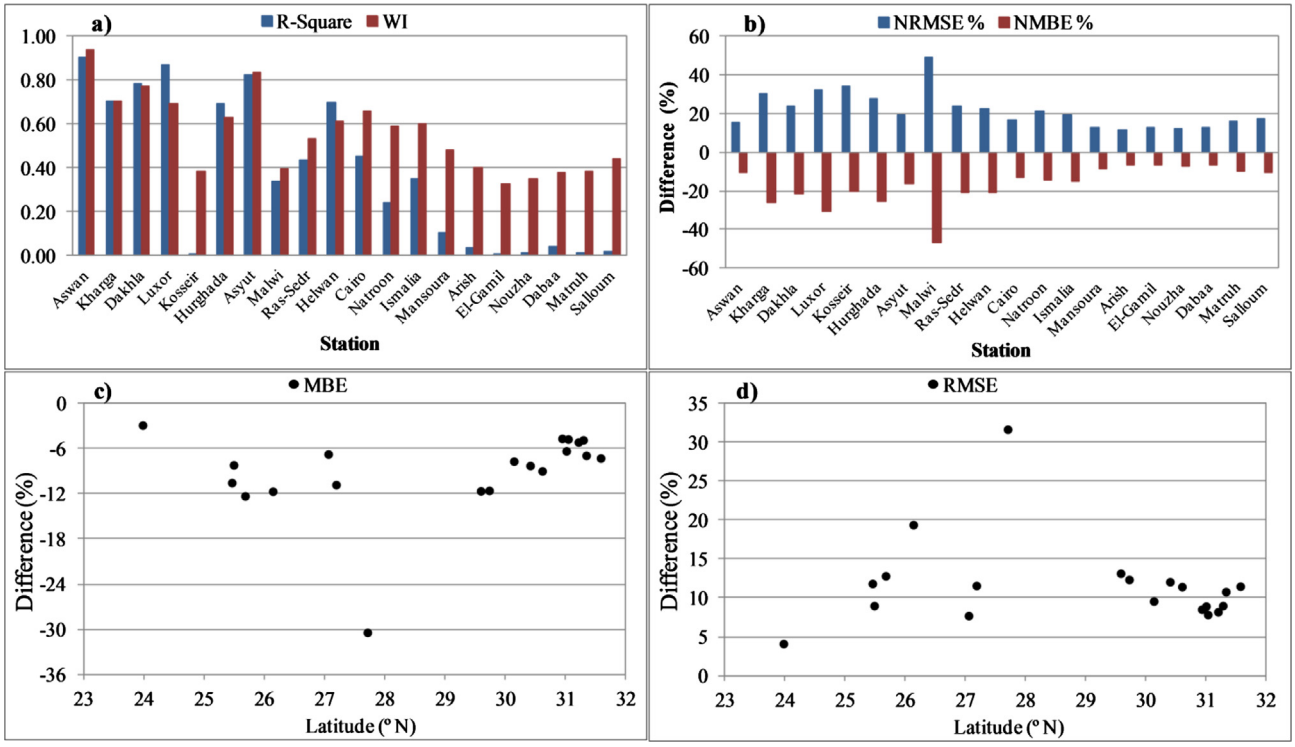


Fig. 7. Statistical indicators for monthly average of relative humidity.

Table 6
Percentage of stations, which have R-Square higher than 0.75 and 0.5 with their correspondence WI, RMSE and the dominant MBE.

Variable	Percentage of stations (%) corresponding to						Percentage of stations corresponding to the dominant MBE
	R-Square	WI	RMSE	R-Square	WI	RMSE	
	>0.75	>0.90	<5 °C	>0.5	>0.80	<6 °C	
T_{max}	20	20	20	55	50	60	90% with difference between −2 and 2
T_{min}	35	50	90	100	95	100	80% with difference between −3.0 and 3.0
T_{mean}	100	100	100	100	100	100	90% with difference between −2 and 2
T_d	75	55	75	100	70	90	85% with difference between −6 and 0
RH	20	5	5	35	10	5	75% with difference between −12.0 and −5.0

of the ground observations and still needs improvement. Monteiro et al. (2018) have reached the same conclusion based on the assessment of NASA POWER satellite based weather system for the Brazilian conditions. This disagreement of relative air humidity estimations occurred in the coastal weather stations; as well as of the air temperature for those inland stations characterized by high elevations according to Negm et al. (2017) in Italy and this was noticeable at the Egyptian stations.

3.7. General statistical indication

The values 0.25, 0.50 and 0.75 for the coefficient of determination (R^2) match weakly, moderately and significantly respectively (Henseler et al., 2009; Hair et al., 2017). It is noticed that R^2 of 0.5 is relatively matched with WI of 0.80 and RMSE less than 6 °C, while R^2 of 0.75 matches with WI of 0.90 and RMSE less than 5 °C. Table 6 illustrates the percentage of stations that correspond to R^2 by more than 0.5 and 0.75 for each chosen atmospheric variable. One may notice that T_{mean} at all stations has R^2 and WI of more than 0.75 and 0.90 respectively, and RMSE less than 5 °C. One may conclude that the reanalysis NASA POWER T_{mean} is highly correlated with the observations compared to other studied variables. On the other hand, the reanalysis data for RH has the lowest accuracy due to little correlation with observation as shown in Table 6. Where, only 20% of the stations have R^2 more than 0.75 that matches with the WI of 0.90 and RMSE less than 5% at only 5% of stations.

NASA POWER is more accurate with goodness of fit at about 100% and 75% of the selected stations that has R^2 more than 0.75 for T_{mean} and T_d respectively. Also, it can only simulates T_{max} at 20% and T_{min} at 35% of stations (has R^2 more than 0.75) because it influenced by the Mediterranean Sea (northern of 30°N and sometimes reaches to 26°N) and western of 30°E with distance between 2.5 and 14.5 km from the Sea with any elevations

(e.g., Salloum, Matruh, Dabaa and Nouzha). On contrary, the best values of the statistical indicators for RH are recorded at only 20% of stations (has R^2 more than 0.75) that not influenced by the Mediterranean Sea (southern of 26°N). Finally, NASA POWER success to simulate temperature throughout the area that dominated by the influence of the Mediterranean Sea (northern of 30°N and may be reaches to 26°N) and vice versa for RH. As well as, the higher the RMSE and MBE values associated to RH the higher the RMSE and MBE associated to all temperature parameters that appear at Malwi station in middle Egypt, while this condition not achieved with the lowest RMSE and MBE.

4. Conclusions

The statistical calculations indicated that the NASA POWER reanalysis data for temperature parameter have a reasonably good fit to the observations at all selected ground stations, especially the northern coastal ones influence by the Mediterranean Sea (northern of 30°N and sometimes reaches to 26°N) and western of 30°E with distance between 2.5 and 14.5 km from the Sea with any elevations (Salloum, Matruh, Dabaa and Nouzha). While the best estimation for RH were recorded at stations that are located south of the area that dominated by the influence of the Mediterranean Sea (southern of 26°N; Aswan, Luxor, Dakhla and Kharga). Additionally, NASA POWER dataset for T_d is acceptable and highly correlated in the middle and southern parts of Egypt compared to T_{max}, T_{min}, and RH. The correlation between NASA POWER reanalysis and the observed data for RH increases gradually southward and has its maximum at Aswan. On the other hand, the NASA POWER reanalysis dataset for T_{mean} are better than other variables (T_{max}, T_{min}, T_d and RH) all over Egypt, where R^2 and WI values at all selected stations are higher than 0.75 and 0.90 associated with RMSE less than 5 °C. The selected meteorological parameters could rank from the highest to the lowest accuracy as T_{mean}, T_{min}, T_d, T_{max} and RH. Generally, it can be concluded that there is a relative degree of consistency between NASA POWER reanalysis datasets and the ground observations over Egypt. This proved that the NASA POWER reanalysis dataset is an acceptable substitute in Egypt in case of missing or availability limitations of observations. However, it still needs improvements by considering the influence of Mediterranean Sea and the locality of (especially Middle; at Malwi station which has the maximum RMSE and MBE for all elements) Egypt particularly relative humidity estimations.

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